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REVIEW ARTICLE

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Long-term outcomes of maxillary single-tooth implants in relation to timing protocols of implant placement and loading: Systematic review and meta-analysis

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Abstract

Objectives: The aim of the present systematic literature review was to determine whether long-term treatment results with single-tooth implants may differ depending on the timing of implant placement in relation to tooth extraction (immediate IP/early EP/delayed DP) and the timing of prosthetic loading (immediate IL/early EL/delayed DL).

Material and Methods: Electronic and manual searches were performed to identify studies reporting on long-term results (survival rate and/or marginal bone resorption after ≥3 years) of maxillary single-tooth implants in the aesthetic zone using defined placement and loading protocols. Comparative trials were subjected to meta-analyses whilst data from single-arm studies were pooled to evaluate differences between timing protocols.

Results: A total of 7 controlled trials were considered for meta-analyses: immediate loading was compared to delayed loading in 3 studies on immediate placement (IPIL vs. IPDL, $p = .306$) and in 2 studies on delayed placement (DPIL vs. DPDL, $p = 1.000$) whilst 2 studies compared early versus delayed placement with delayed loading (EPDL vs. DPDL, $p = .600$), however, without significant differences. Pooled data analysis of 29 studies (965 implants) did not show differences between timing of placement or loading as well as marginal bone remodelling. No impact of the one abutment – one time concept, flap design and simultaneous bone or soft tissue augmentation could be established.

Conclusions: Insufficient data are available for meta-analytic comparison of all combinations of implant placement and loading protocols.

KEYWORDS
biological outcomes, bone level change, dental implant, implant loading protocols, implant placement protocols, success rate, survival rate

1 | INTRODUCTION

Single-tooth replacement in the aesthetic zone of the maxilla is amongst the most complex challenges in dental implantology. This is partly due to the history of tooth loss that may involve previous root resection surgery or traumatic injuries and the special anatomic situation for implant placement (Farmer & Darby, 2014). Furthermore, biologic concerns are highly relevant for successful treatment outcomes but also the aesthetic results that rarely go unnoticed by the patients who present with increasing demands (Hof et al., 2014). Therapeutic success is nowadays measured by means of patient satisfaction and aesthetic indices, such as the Pink Esthetic Score (Fürhauser et al., 2005), and however, long-term implant survival and biologic success in terms of marginal bone stability still provide the foundation for successful outcomes. Time-saving approaches regarding both the immediate or early insertion of implants into the socket after tooth removal as well as immediate or early rehabilitation using fixed implant-supported (provisional) crowns have been developed on account of obvious benefits for patients and doctors (Pommer et al., 2014). Simultaneous surgical interventions such as guided bone regeneration (Wessing et al., 2017), soft tissue grafting (Zuiderveld et al., 2018) and guided flapless implant placement (Hingsammer et al., 2018) further expand the diversity and complexity of single-tooth implant rehabilitation in the aesthetic zone. In recent years, the 'one abutment-one time' concept has gained popularity (Fürhauser et al., 2017) due to potential advantages of immediate insertion of final (zirconia) abutments never to be removed thereafter (Becker et al., 2012).

Several systematic reviews have investigated early complications following single-tooth replacement comparing immediate versus early versus delayed implant placement (Chen & Buser, 2014; Cosyn et al., 2019) as well as immediate versus early versus delayed implant loading protocols (Benic et al., 2014; Pigozzo et al., 2018). Single-tooth implants may be particularly susceptible to early failures within the first months of osseointegration both due to instant insertion into unhealed bone (Canellas et al., 2019) and due to premature overload (Grandi et al., 2015), and however, evidence on mid- to long-term success is yet very limited. Furthermore, interactive effects of various placement and loading protocols have been largely disregarded so far. Therefore, the aim of the present review was to analyse all 9 available combinations of placement and loading protocols (IPIL, EPIL, DPIL, IPEL, EPEL, DPEL, IPDL, EPDL and DPDL) regarding long-term survival and marginal bone resorption around single-tooth implants in the aesthetic zone.

2 | MATERIALS AND METHODS

The present systematic review was performed in due consideration of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) proposed by Moher et al. (2010).

2.1 | Objectives

The focussed question of the present review was, do timing-based implant treatment protocols influence the biologic outcomes of single-tooth implants and restorations in the aesthetic zone? The primary objective was to compare timing-based implant treatment concepts of single-tooth gaps regarding long-term biologic outcome. The PICOS elements relating to this focussed research question were as follows: *Patients* =adults in need of a single-tooth implant in the aesthetic zone (upper incisors and canines), *Intervention* =immediate/early implants subjected to immediate/early loading, *Comparison* =delayed implants subjected to delayed loading, *Outcome* =implant survival and marginal bone resorption after ≥3 years, *Study designs* =randomised controlled trials and prospective or retrospective clinical studies or case series. The secondary objective was to investigate the impact of immediate abutment placement (one abutment-one time concept) on the same outcomes.

2.2 | Inclusion criteria

The criteria for inclusion comprised: (1) prospective or retrospective clinical studies or case series, (2) at least 10 patients with single-tooth implants, (3) follow-up period of at least 3 years, (4) detailed information on timing of implant placement and timing of prosthetic rehabilitation, (5) outcomes in the anterior maxilla reported in terms of implant survival and/or marginal bone resorption, (6) two-piece titanium or ceramic implants restored with abutments of any material, (7) with or without bone or soft tissue augmentation procedures and (8) articles written in English language. Reasons for exclusion involved insufficient data (relevant to subgroups of interest), case reports, animal studies and review publications.

2.3 | Search strategy and risk of bias assessment

An electronic literature search of the time period January 2000 to April 2020 was performed in PubMed, EMBASE and Cochrane Library databases using the following search terms: (dental implant) and (immediate OR early OR late OR delayed OR conventional OR post-extraction OR post-extractive). The filters were set to English language, Humans and the stated time period. Articles that qualified at the title and abstract level were evaluated by two researchers (M.D. and B.P.) in full text. Inter-rater reliability in the selection of appropriate studies was evaluated by calculating kappa coefficients. In addition, hand searches were performed on the reference lists of all included articles and relevant review publications by two researchers (M.D. and B.P.) independently. Any disagreement that arose at full-text level was resolved by discussion with a third reviewer (R.H.). When multiple publications on the same study with different follow-up were encountered, the paper with the longest follow-up was included.

The same two reviewers extracted all relevant data from the final selection of articles (Table 1) and assessed the risk of bias for controlled trials as suggested in the Cochrane Handbook for Systematic Reviews of Interventions version 6.1 chapter 8 (Higgins et al., 2020) via the following criteria: random sequence generation (selection bias), allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting and other sources of bias. Cohen's Kappa was calculated to assess inter-rater reliability in quality assessment.

2.4 | Statistical analysis

Meta-analysis for the dichotomous variable implant survival was performed using a random effect model (package epiR), and effect sizes were expressed as risk ratios (RRs) and 95% confidence intervals ($^{95\%}$CIs). Drop-outs were excluded from the analysis altogether. Meta-analysis of pooled data was performed using random effect inverse variance modelling with DerSimonian–Laird estimator and continuity correction of 0.5 in studies with zero cell frequencies (package metaprop). Data on marginal bone remodelling were pooled across studies using weighted mean values and compared between subgroups using Mann–Whitney U-tests. Differences in mean time of follow-up were also investigated via Mann–Whitney U-tests. The level of significance was set at $p = .05$. All analyses were performed using R Project software (R Foundation for Statistical Computing, Vienna, Austria).

3 | RESULTS

3.1 | Search results

The search strategy yielded a total of 7029 publications, of which 6786 entered title and abstract screening (Figure 1). A total of 121 studies underwent full-text screening and data extraction, corresponding authors were contacted for additional data. The final selection constituted 29 articles (Table 1) of which 7 represented controlled trials comparing 2 treatment modalities and thus were suitable for meta-analytic evaluation and risk of bias assessment. Kappa statistics were 0.871 ($p < .001$) at the title level, 0.857 ($p < .001$) at the abstract level and 0.922 ($p < .001$) in quality assessment (Table 2) indicative of substantial agreement.

In the included studies, immediate placement was defined as implant insertion at the day of tooth removal, early placement as 10 days to 8 weeks after extraction and delayed placement as 3–6 months thereafter. Immediate provisional restoration was performed at the same day, early loading after 3–8 weeks and delayed loading after 3–6 months, respectively. Only 2 studies (Jemt, 2008; Turkyilmaz et al., 2007) reported on machined surfaces whilst the majority used rough surfaced implants.

3.2 | Meta-analysis of comparative trials

The first meta-analytic comparison was possible between immediate implants subjected to immediate versus delayed loading (IPIL vs. IPDL). Three studies were included (Arora & Ivanovski, 2018; Bell & Bell, 2014; Slagter et al., 2021) and two showed high risk of bias in 2–3 criteria. Overall implant survival rates were 96.2% with IPIL versus 98.3% with IPDL (Figure 2), however, did not differ significantly (RR = 2.14 [$^{95\%}$CI 0.49–9.35], $p = .306$). Two studies reported on marginal bone resorption (0.35 vs. 0.37 mm, $p = .667$) after a follow-up of 3–5 years.

The second meta-analysis was performed to compare delayed implants with either immediate or delayed loading (DPIL vs. DPDL). Both studies included (Degidi et al., 2009, den Hartog et al., 2016) were randomised controlled trials with low or unclear risk of bias in all criteria. Overall implant survival rates were 98.1% in both groups (Figure 3, RR = 1 [$^{95\%}$CI 0.11–9.26], $p = 1.000$), and no marginal bone loss analysis could be performed as reported in one study only (0.85 vs. 0.75 mm after 3 years). Follow-up was heterogeneous as the other study reported results after 5 years.

The third meta-analytic comparison was between early versus delayed implant placement both subjected to delayed loading (EPDL vs. DPDL). Two studies were included (Gotfredsen, 2012 controlled trail, Schropp & Isidor, 2008 randomised controlled trial) and one showed high risk of bias in 2 criteria. Overall implant survival rates were 95% with EPDL versus 100% with DPDL (Figure 4), however, did not differ significantly (RR =1.9 [$^{95\%}$CI 0.17–21.07], $p =.600$), and no marginal bone loss data were available. Both studies had a follow-up of 5 years.

3.3 | Pooled data analysis across studies

Data on 965 implants were extracted from 29 studies (Table 1) and evaluated regarding the combination of implant placement and implant loading protocols: implant survival rates were 99.6% for 414 IPIL implants [$^{95\%}$CI 98.5–100], 97.9% for 203 IPDL implants [$^{95\%}$CI 95.6–100], 100% for 34 EPEL implants [$^{95\%}$CI 94.5–100], 95.8% for 20 EPDL implants [$^{95\%}$CI 85.0–100], 99.1% for 104 DPIL implants [$^{95\%}$CI 96.0–100], 92.4% for 52 DPEL implants [$^{95\%}$CI 85.4–99.9] and 100% for 138 DPDL implants [$^{95\%}$CI 97.6–100]. No difference could be seen between protocols for implant placement. When comparing implant loading modalities, there was no difference between immediate (99.5% [$^{95\%}$CI 98.5–100], early (97.3% [$^{95\%}$CI 92.9–100]) and delayed loading (98.9% [$^{95\%}$CI 97.2–100]).

Data on marginal bone resorption were available for 8 studies on IPIL (145 implants), two studies on IPDL (34 implants), two studies on DPIL (51 implants) and 5 studies on DPDL (110 implants). No difference between immediate versus delayed implant placement (0.5 vs. 1.1 mm) and between immediate versus delayed implant loading (0.6 vs. 1.1 mm) could be substantiated, nor between any of the subgroups.

TABLE 1 Details on 29 included studies reporting on 965 implants

Study (author, year)	Study design	Follow-up (months)	Implants placed	Implant drop-outs	Implants lost	Mean age (years)	Flap retraction	Bone graft surgery	Soft tissue graft	Implant brand
Immediate Placement +Immediate Loading (IPIL)										
Mijiritsky et al. (2009)	retrospective	40.7	20	0	0	42.0	no	yes	n.d.	Xive, Frialit–2
Paul & Held (2013)	retrospective	60	8	2	0	44.8	n.d.	yes	yes	Nobel Biocare
Bell and Bell (2014)	retrospective	36	42	0	3	56.3	yes	yes	no	Nobel Biocare
Berberi et al. (2014)	retrospective	60	18	0	1	n.d.	n.d.	no	n.d.	Astra Tech
Calvo-Guirado et al. (2014)	prospective	120	61	0	0	39.6	n.d.	no	n.d.	Biomet 3i
Calvo-Guirado et al. (2015)	prospective	36	50	0	0	37.5	n.d.	no	n.d.	MIS Implants
Crespi et al. (2015)	retrospective	36	63	0	0	52.4	no	n.d.	n.d.	Ticino Forniture Dentali
Guarnieri et al. (2016)	retrospective	36	10	0	0	42.0	no	no	n.d.	BioHorizons
Nölken et al. (2016)	prospective	36	18	1	1	40.9	no	yes	no	Astra Tech
Van Nimwegen et al. (2016)	retrospective	48	48	13	2	48.0	no	yes	no	Zimmer Biomet
Arora et al. (2017)	prospective	60	30	0	0	n.d.	no	yes	no	Astra Tech
Arora & Ivanovski (2018)	retrospective	36	20	0	0	44.7	no	yes	n.d.	Astra Tech
Nölken et al., 2018	retrospective	45	26	0	0	48.4	no	yes	yes	Astra Tech
Slagter et al. (2021)	RCT	60	18	2	0	39.0	yes	yes	no	Nobel Biocare
Immediate Placement +Delayed Loading (IPDL)										
Covani et al. (2012)	prospective	120	36	0	2	n.d.	yes	yes	n.d.	Sweden&Martina
Bell and Bell (2014)	retrospective	36	84	0	2	60.9	yes	yes	no	Nobel Biocare
Montoya-Salazar et al. (2014)	prospective	36	18	0	0	n.d.	yes	yes	n.d.	MIS Implants
Cecchinato et al. (2015)	prospective	36	31	0	1	51.0	n.d.	n.d.	n.d.	Astra Tech
Arora and Ivanovski (2018)	retrospective	36	20	0	0	56.4	yes	yes	n.d.	Astra Tech
Slagter et al. (2021)	RCT	60	17	3	0	42.0	yes	yes	yes	Nobel Biocare
Early Placement +Early Loading (EPEL)										
Buser et al., 2008	retrospective	36	16	0	0	39.9	yes	yes	n.d.	Straumann
Chappuis et al. (2018)	prospective	120	18	0	0	41.7	yes	yes	no	Straumann

(Continues)

TABLE 1 (Continued)

Study (author, year)	Study design	Follow-up (months)	Implants placed	Implant drop-outs	Implants lost	Mean age (years)	Flap retraction	Bone graft surgery	Soft tissue graft	Implant brand
Early Placement +Delayed Loading (EPDL)										
Schropp and Isidor (2008)	RCT	60	12	0	1	47.0	yes	no	n.d.	Biomet 3i
Gotfredsen (2012)	prospective	120	8	0	0	35.0	yes	yes	n.d.	Astra Tech
Delayed Placement +Immediate Loading (DPIL)										
Degidi et al. (2009)	RCT	36	30	0	0	31.5	yes	no	no	Xive
Zafiropoulos et al. (2010)	retrospective	60	18	0	0	51.2	yes	yes	no	Camlog, Straumann
Berberi et al. (2014)	retrospective	60	17	0	1	n.d.	yes	no	n.d.	Astra Tech
den Hartog et al. (2016)	RCT	60	24	3	1	38.4	yes	yes	no	Nobel Biocare
Raes et al. (2018)	prospective	96	18	0	0	42.0	yes	n.d.	n.d.	Astra Tech
Delayed Placement +Early Loading (DPEL)										
Cooper et al. (2007)	prospective	36	43	6	3	30.6	n.d.	n.d.	n.d.	Astra Tech
Turkyilmaz et al. (2007)	prospective	48	15	0	1	40.0	n.d.	no	n.d.	Brånemark
Delayed Placement +Delayed Loading (DPDL)										
Turkyilmaz et al. (2007)	prospective	48	9	0	0	40.0	n.d.	n.d.	n.d.	Brånemark
Jemt (2008)	retrospective	180	47	10	0	25.4	yes	no	n.d.	Brånemark
Schropp and Isidor (2008)	RCT	60	10	0	0	47.0	yes	yes	n.d.	Biomet 3i
Degidi et al. (2009)	RCT	36	30	0	0	31.5	yes	no	no	Xive
Gotfredsen (2012)	prospective	120	10	0	0	31.0	yes	no	n.d.	Astra Tech
den Hartog et al. (2016)	RCT	60	26	3	0	40.1	yes	yes	no	Nobel Biocare
Guarnieri et al. (2016)	retrospective	36	9	0	0	40.0	yes	yes	no	BioHorizons
Canullo et al. (2018)	prospective	60	22	12	0	68.3	yes	yes	no	Sweden&Martina

Abbreviations: n.d, no data; RCT, randomized controlled trial.

FIGURE 1 Flow chart

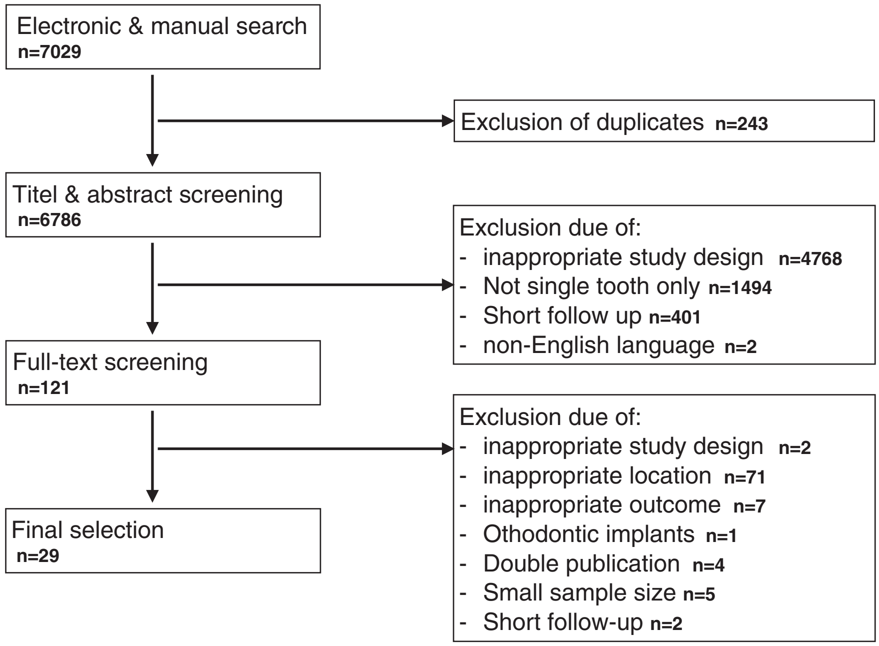

TABLE 2 Risk of bias assessment of controlled trials regarding selection bias

Study	RSG	AC	PB	DB	AB	SRB	OSB
Arora & Ivanovski (2018)	High	High	Low	Low	Low	Unclear	Low
Bell and Bell (2014)	High	High	Low	Low	Low	High	Low
Degidi et al. (2009)	Low	Low	Low	Unclear	Low	Low	Low
den Hartog et al. (2016)	Low	Low	Low	Low	Unclear	Low	Low
Gotfredsen (2012)	High	High	Unclear	Low	Unclear	Low	Low
Schropp and Isidor (2008)	Unclear	Unclear	Unclear	Low	Unclear	Low	Low
Slagter et al. (2021)	Low	Low	Low	Unclear	Low	Low	Low

Abbreviations: AB, attrition bias; AC, allocation concealment; DB, detection bias; OSB, other sources of bias; PB, performance bias; RSG, random sequence generation; SRB, selective reporting bias.

Information on flap retraction was available in 22 studies (738 implants), and no difference in the survival of flapped versus flapless implants could be found (98.1% vs. 98.6%), yet marginal bone resorption was slightly higher (0.65 vs. 0.13 mm) for implants placed with a flap. Data on bone graft surgery were reported in 25 studies (802 implants), and however, there was no difference after immediate, early and delayed placement. No impact of simultaneous bone grafting on immediate (0.36 vs. 0.85 mm) or delayed placement (0.35 vs. 1.40 mm) was observed. Information on simultaneous soft tissue grafting was available in 12 studies (429 implants), and no differences were observed between the groups.

3.4 | One abutment-one time concept

The impact of immediate abutment placement could be analysed in 18 studies on immediate implant loading (518 implants, of which 53 received immediate final abutments). The one abutment-one time concept, however, did not affect implant survival rates.

4 | DISCUSSION

The objective of the present systematic review was to compare studies reporting follow-up data at least 3 years after single-tooth implant placement in the canine-to-canine region of the maxilla loaded at different time intervals. The results in terms of aesthetics are part of a separate meta-analysis. The major problem of this investigation was missing data, despite the fact that not only RCTs but also studies of lower evidence levels were included. In this context, further evaluation of soft tissue parameters (peri-implant pocket depth, bleeding on probing, shape of keratinized mucosa) was not possible. Meta-analytic comparison between 2 groups within the same studies was possible only for 3 combinations of implant placement and loading protocols (IPIL vs. IPDL, DPIL vs. DPDL and EPDL vs. DPDL), however, did not produce any significant findings regarding both survival rate and marginal bone remodelling. In fact, the level of evidence (based on the GRADE classification) was judged as low or very low in all 3 cases (Table 3).

	Immediate loading		Delayed loading			Risk Ratio	
Study	Total	Events	Total	Events	Weight	random-effect	95% CI
Arora & Ivanovski 2018	20	0	20	0	21.64%	1	[0.02 48.09]
Bell & Bell 2014	42	3	84	2	56.39 %	3	[0.52 17.27]
Slagter et al. 2021	16	0	14	0	21.97%	0.88	[0.02 41.80]
Total	78		118		100,00 %	2.14	[0.49 9.35]
Total Events		3		2			

Heterogeneity: Chi2=0.49, df=2, p-value 0.781, I2=100%

Test for overall effect: Z=1.0226, p-value 0.306

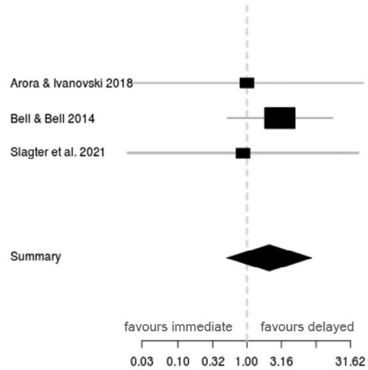

FIGURE 2 Meta-analytic comparison between immediate implants subjected to immediate versus delayed loading (IPIL vs. IPDL)

	Immediate loading		Delayed loading			Risk Ratio	
Study	Total	Events	Total	Events	Weight	random-effect	95% CI
Degidi et al. 2009	30	0	30	0	33.33%	1	[0.02 48.82]
den Hartog et al. 2016	24	1	24	1	66.67%	1	[0.07 15.08]
Total	54		54		100,00 %	1	[0.11 9.26]
Total Events		1		1			

Heterogeneity: Chi2=0 , df=1, p-value 1, I2=100%

Test for overall effect: Z=0, p-value 1

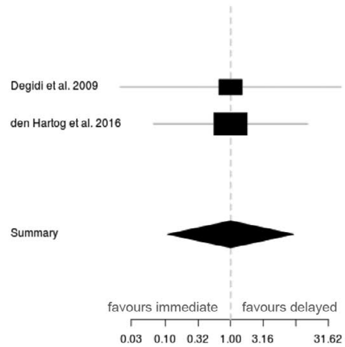

FIGURE 3 Meta-analytic comparison between delayed implants subjected to immediate versus delayed loading (DPIL vs. DPDL)

	Early placement		Delayed placement		Weight	Risk Ratio	
Study	Total	Events	Total			random-effect	95% CI
Gotfredsen 2012	8	0	10	0	39.67%	1.22	[0.03 55.72]
Schropp & Isidor 2008	12	1	10	0	60.33%	2.54	[0.11 56.25]
Total	20		20		100,00 %	1.9	[0.17 21.07]
Total Events		1		0			

Heterogeneity: Chi2=0.085 , df=1, p-value 0.77, I2=100%

Test for overall effect: Z=0.523, p-value 0.60

FIGURE 4 Meta-analytic comparison between early versus delayed implant placement subjected to delayed loading (EPDL vs. DPDL)

Analysis of data pooled across studies was feasible for only 7 of the 9 possible combinations of placement and loading modalities as no published long-term data on immediate placement with early loading (IPEL) and early placement with immediate loading (EPIL) could be found. No differences could be substantiated between implant placement protocols regarding implant survival and long-term marginal bone resorption. It has to be admitted, that in 1 out of 2 studies in the DPEL group and 2 out of 8 studies in the DPDL group implants with machined surfaces were used, whereas implants with a rough surface were investigated in all other subgroups. Thus, no conclusion can be drawn on the long-term results of immediately or early placed implants with a machined surface.

As the length of follow-up was not homogeneous between the studies pooled together (range: 36–120 months), we considered the risk of a biased comparison as a result of different follow-up times. However, no statistical difference could be found between the follow-up in the immediate versus early loading groups (54.8 vs. 55.7 months, $p = .648$), nor between the IPEL and DPEL groups (53.6 vs. 39.5 months, $p = .598$). As information arising from between-study comparison are associated with increased potential for bias and confounding (Deeks et al. 2003), the present pooled analysis, however, needs to be interpreted with great care.

Our results are inconsistent with those of a recent systematic review and meta-analysis (Cosyn et al., 2019) that found significantly

TABLE 3 Summary of findings (GRADE)

Patients: adults in need of a single-tooth implant in the aesthetic zone (upper incisors/canines)
Intervention: immediate/early implants subjected to immediate/early loading
Comparison: delayed implants subjected to delayed loading

Outcome	Survival rates		Relative risks (95% CI)	No of patients (studies)	GRADE	Comments
Implant survival after 3–5 years (immediate placement)	immediate loading 96.2%	delayed loading 98.3%	2.1 (0.5–8.7) $p = .306$	196 (3)	+ooo very low[a]	relevant to rough surfaces only
Implant survival after 5 years (delayed placement)	immediate loading 98.1%	delayed loading 98.1%	(0.1–9.2) $p = 1.000$	108 (2)	++oo Low[b]	relevant to rough surfaces only
Implant survival after 5 years (delayed loading)	early placement 95.0%	delayed placement 100%	1.9 (0.1–20.9) $p = .584$	40 (2)	+ooo very low[a]	relevant to rough surfaces only

[a]Graded down −1 due to imprecision.
[b]Graded down −1 due to risk of bias and −1 due to imprecision.

more early failures after immediate implant placement compared to delayed implant placement (94.9% vs. 98.9%). Short-term results, however, can not necessarily be extrapolated to mid- and long-term outcomes. Another meta-analysis of 10 RCTs comparing implant survival and marginal bone loss after 1 year of loading (Benic et al., 2014) could not find significant differences between immediate and delayed loading protocols and it thus in line with our findings. A recent meta-analytic comparison on immediate versus early loading modalities (Pigozzo et al., 2018), however, found no differences regarding implant survival and marginal bone resorption after 1–3 years. Our review, by contrast, showed a lower long-term survival rate for single-tooth implants subjected to early occlusal loading compared to immediate provisional restoration. In a Cochrane review on loading protocols (Esposito et al., 2013), 15 RCTs comparing immediate versus delayed loading and 6 RCTs comparing immediate versus early loading were included, however, studies were not limited to single-tooth implants in the aesthetic zone. No evidence of differences in implant failures was found, but some evidence of a small reduction in marginal bone loss favouring immediate loading.

Most of the systematic reviews on single-tooth implants in the aesthetic zone also included first or even second premolars, and however, research seems to indicate that bone remodelling patterns differs substantially in premolar areas compared to incisors and canines. This is due to the fact that in more than 50% the buccal alveolar bone in the anterior region is less than 0.5 mm thick (Cho et al., 2019; Huynh-Ba et al., 2010; Januário et al., 2011), especially at the level of the alveolar crest (Vera et al., 2012) relevant for marginal bone stability. Furthermore, maxillary incisor and canine roots are located at the buccal fifth of the alveolar crest (Kim et al., 2011) supporting the assumption that different bone remodelling patterns around immediate implants may occur. A considerable number of published papers could not be included in this review because they reported on the combined results of anterior and premolar implants and the data could not be attributed to the canine-to-canine region of the maxilla as was the topic of this review.

No difference between flap retraction and flapless implant placement was found in the present meta-analysis (based on pooled data of 738 implants) which is in line with the same finding of a recent systematic review (Lin et al., 2014). However, it has to be considered that this review included both maxillary and mandibular implants placed into extraction sockets as well as healed ridges in a variety of different indications. Details on the aesthetic outcome were not investigated in the present meta-analysis as another group of the Consensus Conference put their focus on aesthetics. Pooled data analysis, however, indicated that simultaneous soft tissue grafting and gap filling around immediate implants does not improve biologic outcomes of single-tooth implants in the aesthetic zone.

Regarding timing definitions, we observed some variability in the criteria used across the studies included: early placement was performed between 10 days and 8 weeks after tooth extraction, and early loading was defined as 3–8 weeks after implant placement. These substantial differences may very likely impact treatment results regarding both survival rates as well as marginal bone remodelling. It seems advisable to follow strict guidelines when evaluating early implant placement and early loading to render comparison between studies possible.

Future research is thus indicated to elucidate the question of preferable combinations of implant placement and loading protocols for single-tooth implants in the aesthetic zone of the maxilla. Randomised controlled trials comparing various treatment modalities within the same study are difficult to carry out without high percentages of dropouts in the long term, however, are needed to gain strong evidence on everyday treatment decisions that are key to achieving biologic and aesthetic success. Special emphasis may be put on strict inclusion criteria concerning both the patients' risk factors and anatomical baseline situation, homogeneity of surgical technique (particularly considering simultaneous bone and soft tissue augmentation procedures) and quality of reporting.

5 | CONCLUSION

Within the limits of the present meta-analysis, it can be concluded that there is insufficient evidence on ≥3-year outcomes of single-tooth implants placed in the anterior region of the maxilla to show differences between protocols regarding times of implant placement and loading protocols. Neither implant survival nor marginal bone level changes could be attributed to implant placement protocols in relation to tooth extraction and immediate, early or delayed implant loading up to a 5-year follow-up.

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AUTHOR CONTRIBUTIONS

B.P., M.D. and R.H. contributed equally to literature search, data extraction and analysis, as well as writing. L.L.A. and J.P. contributed to literature search.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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